

52. $z = 10 - x^2 - y^2$, $z = 1$
 53. $z = x^2 + y^2$, $x^2 + y^2 = 25$, $z = 0$
 54. $y = x^2 + z^2$, $2y = x^2 + z^2 + 4$
 55. Find the centroid of the homogeneous solid that is bounded by the hemisphere $z = \sqrt{a^2 - x^2 - y^2}$ and the plane $z = 0$.
 56. Find the center of mass of the solid that is bounded by the graphs of $y^2 + z^2 = 16$, $x = 0$, and $x = 5$ if the density at a point P is directly proportional to distance from the yz -plane.
 57. Find the moment of inertia about the z -axis of the solid that is bounded above by the hemisphere $z = \sqrt{9 - x^2 - y^2}$ and below by the plane $z = 2$ if the density at a point P is inversely proportional to the square of the distance from the z -axis.
 58. Find the moment of inertia about the x -axis of the solid that is bounded by the cone $z = \sqrt{x^2 - y^2}$ and the plane $z = 1$ if the density at a point P is directly proportional to the distance from the z -axis.

In Problems 59–62, convert the point given in spherical coordinates to (a) rectangular coordinates and (b) cylindrical coordinates.

59. $\left(\frac{2}{3}, \frac{\pi}{2}, \frac{\pi}{6}\right)$ 60. $\left(5, \frac{5\pi}{4}, \frac{2\pi}{3}\right)$
 61. $\left(8, \frac{\pi}{4}, \frac{3\pi}{4}\right)$ 62. $\left(\frac{1}{3}, \frac{5\pi}{3}, \frac{\pi}{6}\right)$

In Problems 63–66, convert the points given in rectangular coordinates to spherical coordinates.

63. $(-5, -5, 0)$ 64. $(1, -\sqrt{3}, 1)$
 65. $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}, 1\right)$ 66. $\left(-\frac{\sqrt{3}}{2}, 0, -\frac{1}{2}\right)$

In Problems 67–70, convert the given equation to spherical coordinates.

67. $x^2 + y^2 + z^2 = 64$ 68. $x^2 + y^2 + z^2 = 4z$
 69. $z^2 = 3x^2 + 3y^2$ 70. $-x^2 - y^2 + z^2 = 1$

In Problems 71–74, convert the given equation to rectangular coordinates.

71. $\rho = 10$ 72. $\phi = \pi/3$
 73. $\rho = 2 \sec \phi$ 74. $\rho \sin^2 \phi = \cos \phi$

In Problems 75–82, use triple integrals and spherical coordinates. In Problems 75–78, find the volume of the solid that is bounded by the graphs of the given equations.

75. $z = \sqrt{x^2 + y^2}$, $x^2 + y^2 + z^2 = 9$
 76. $x^2 + y^2 + z^2 = 4$, $y = x$, $y = \sqrt{3}x$, $z = 0$, first octant
 77. $z^2 = 3x^2 + 3y^2$, $x = 0$, $y = 0$, $z = 2$, first octant
 78. Inside $x^2 + y^2 + z^2 = 1$ and outside $z^2 = x^2 + y^2$
 79. Find the centroid of the homogeneous solid that is bounded by the cone $z = \sqrt{x^2 + y^2}$ and the sphere $x^2 + y^2 + z^2 = 2z$.
 80. Find the center of mass of the solid that is bounded by the hemisphere $z = \sqrt{1 - x^2 - y^2}$ and the plane $z = 0$ if the density at a point P is directly proportional to the distance from the xy -plane.
 81. Find the mass of the solid that is bounded above by the hemisphere $z = \sqrt{25 - x^2 - y^2}$ and below by the plane $z = 4$ if the density at a point P is inversely proportional to the distance from the origin. [Hint: Express the upper ϕ limit of integration as an inverse cosine.]
 82. Find the moment of inertia about the z -axis of the solid that is bounded by the sphere $x^2 + y^2 + z^2 = a^2$ if the density at a point P is directly proportional to the distance from the origin.

9.16 Divergence Theorem

Introduction In Section 9.14 we saw that Stokes' theorem was a three-dimensional generalization of a vector form of Green's theorem. In this section we present a second vector form of Green's theorem and its three-dimensional analogue.

Another Vector Form of Green's Theorem Let $\mathbf{F}(x, y) = P(x, y)\mathbf{i} + Q(x, y)\mathbf{j}$ be a two-dimensional vector field, and let $\mathbf{T} = (dx/ds)\mathbf{i} + (dy/ds)\mathbf{j}$ be a *unit tangent* to a simple closed plane curve C . In (1) of Section 9.14 we saw that $\oint_C (\mathbf{F} \cdot \mathbf{T}) ds$ can be evaluated by a double integral involving curl $\mathbf{F}$. Similarly, if $\mathbf{n} = (dy/ds)\mathbf{i} - (dx/ds)\mathbf{j}$ is a *unit normal* to C (check $\mathbf{T} \cdot \mathbf{n}$), then $\oint_C (\mathbf{F} \cdot \mathbf{n}) ds$ can be expressed in terms of a double integral of $\text{div } \mathbf{F}$. From Green's theorem,

$$\oint_C (\mathbf{F} \cdot \mathbf{n}) ds = \oint_C P dy - Q dx = \iint_R \left[\frac{\partial P}{\partial x} - \left(-\frac{\partial Q}{\partial y} \right) \right] dA = \iint_R \left[\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right] dA;$$

that is,

$$\oint_C (\mathbf{F} \cdot \mathbf{n}) ds = \iint_R \text{div } \mathbf{F} dA. \quad (1)$$

The result in (1) is a special case of the **divergence** or **Gauss' theorem**. The following is a generalization of (1) to 3-space:

Theorem 9.16.1 Divergence Theorem

Let D be a closed and bounded region in 3-space with a piecewise-smooth boundary S that is oriented outward. Let $\mathbf{F}(x, y, z) = P(x, y, z)\mathbf{i} + Q(x, y, z)\mathbf{j} + R(x, y, z)\mathbf{k}$ be a vector field for which P , Q , and R are continuous and have continuous first partial derivatives in a region of 3-space containing D . Then

$$\iint_S (\mathbf{F} \cdot \mathbf{n}) dS = \iiint_D \operatorname{div} \mathbf{F} dV. \quad (2)$$

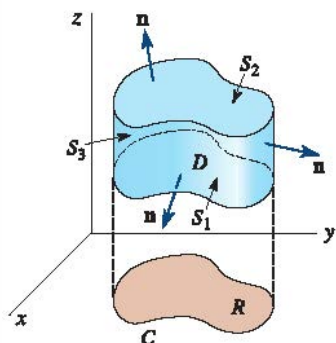


FIGURE 9.16.1 Region D used in proof of Theorem 9.16.1

PARTIAL PROOF: We will prove (2) for the special region D shown in **FIGURE 9.16.1** whose surface S consists of three pieces:

$$\begin{aligned} \text{(bottom)} \quad S_1: z &= f_1(x, y), \quad (x, y) \text{ in } R \\ \text{(top)} \quad S_2: z &= f_2(x, y), \quad (x, y) \text{ in } R \\ \text{(side)} \quad S_3: f_1(x, y) &\leq z \leq f_2(x, y), \quad (x, y) \text{ on } C, \end{aligned}$$

where R is the projection of D onto the xy -plane and C is the boundary of R . Since

$$\operatorname{div} \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \quad \text{and} \quad \mathbf{F} \cdot \mathbf{n} = P(\mathbf{i} \cdot \mathbf{n}) + Q(\mathbf{j} \cdot \mathbf{n}) + R(\mathbf{k} \cdot \mathbf{n}),$$

we can write

$$\iint_S (\mathbf{F} \cdot \mathbf{n}) dS = \iint_S P(\mathbf{i} \cdot \mathbf{n}) dS + \iint_S Q(\mathbf{j} \cdot \mathbf{n}) dS + \iint_S R(\mathbf{k} \cdot \mathbf{n}) dS$$

$$\text{and} \quad \iiint_D \operatorname{div} \mathbf{F} dV = \iiint_D \frac{\partial P}{\partial x} dV + \iiint_D \frac{\partial Q}{\partial y} dV + \iiint_D \frac{\partial R}{\partial z} dV.$$

To prove (2) we need only establish that

$$\iint_S P(\mathbf{i} \cdot \mathbf{n}) dS = \iiint_D \frac{\partial P}{\partial x} dV \quad (3)$$

$$\iint_S Q(\mathbf{j} \cdot \mathbf{n}) dS = \iiint_D \frac{\partial Q}{\partial y} dV \quad (4)$$

$$\iint_S R(\mathbf{k} \cdot \mathbf{n}) dS = \iiint_D \frac{\partial R}{\partial z} dV. \quad (5)$$

Indeed, we shall prove only (5), since the proofs of (3) and (4) follow in a similar manner. Now,

$$\iiint_D \frac{\partial R}{\partial z} dV = \iint_R \left[\int_{f_1(x, y)}^{f_2(x, y)} \frac{\partial R}{\partial z} dz \right] dA = \iint_R [R(x, y, f_2(x, y)) - R(x, y, f_1(x, y))] dA. \quad (6)$$

Next we write

$$\iint_S R(\mathbf{k} \cdot \mathbf{n}) \, dS = \iint_{S_1} R(\mathbf{k} \cdot \mathbf{n}) \, dS + \iint_{S_2} R(\mathbf{k} \cdot \mathbf{n}) \, dS + \iint_{S_3} R(\mathbf{k} \cdot \mathbf{n}) \, dS.$$

On S_1 : Since the outward normal points downward, we describe the surface as $g(x, y, z) = f_1(x, y) - z = 0$. Thus,

$$\mathbf{n} = \frac{\frac{\partial f_1}{\partial x} \mathbf{i} + \frac{\partial f_1}{\partial y} \mathbf{j} - \mathbf{k}}{\sqrt{1 + \left(\frac{\partial f_1}{\partial x}\right)^2 + \left(\frac{\partial f_1}{\partial y}\right)^2}} \quad \text{so that} \quad \mathbf{k} \cdot \mathbf{n} = \frac{-1}{\sqrt{1 + \left(\frac{\partial f_1}{\partial x}\right)^2 + \left(\frac{\partial f_1}{\partial y}\right)^2}}.$$

From the definition of dS we then have

$$\iint_{S_1} R(\mathbf{k} \cdot \mathbf{n}) \, dS = - \iint_R R(x, y, f_1(x, y)) \, dA. \quad (7)$$

On S_2 : The outward normal points upward, so

$$\mathbf{n} = \frac{\frac{\partial f_2}{\partial x} \mathbf{i} + \frac{\partial f_2}{\partial y} \mathbf{j} + \mathbf{k}}{\sqrt{1 + \left(\frac{\partial f_2}{\partial x}\right)^2 + \left(\frac{\partial f_2}{\partial y}\right)^2}} \quad \text{so that} \quad \mathbf{k} \cdot \mathbf{n} = \frac{1}{\sqrt{1 + \left(\frac{\partial f_2}{\partial x}\right)^2 + \left(\frac{\partial f_2}{\partial y}\right)^2}}$$

from which we find

$$\iint_{S_2} R(\mathbf{k} \cdot \mathbf{n}) \, dS = \iint_R R(x, y, f_2(x, y)) \, dA. \quad (8)$$

On S_3 : Since this side is vertical, $\mathbf{k}$ is perpendicular to $\mathbf{n}$. Consequently, $\mathbf{k} \cdot \mathbf{n} = 0$ and

$$\iint_{S_3} R(\mathbf{k} \cdot \mathbf{n}) \, dS = 0. \quad (9)$$

Finally, adding (7), (8), and (9), we get

$$\iint_R [R(x, y, f_2(x, y)) - R(x, y, f_1(x, y))] \, dA,$$

which is the same as (6).

Although we proved (2) for a special region D that has a vertical side, we note that this type of region is not required in Theorem 9.16.1. A region D with no vertical side is illustrated in **FIGURE 9.16.2**; a region bounded by a sphere or an ellipsoid also does not have a vertical side. The divergence theorem also holds for the region D bounded between two closed surfaces, such as the concentric spheres S_a and S_b shown in **FIGURE 9.16.3**; the boundary surface S of D is the union of S_a and S_b . In this case $\iint_S (\mathbf{F} \cdot \mathbf{n}) \, dS = \iiint_D \operatorname{div} \mathbf{F} \, dV$ becomes

$$\iint_{S_b} (\mathbf{F} \cdot \mathbf{n}) \, dS + \iint_{S_a} (\mathbf{F} \cdot \mathbf{n}) \, dS = \iiint_D \operatorname{div} \mathbf{F} \, dV,$$

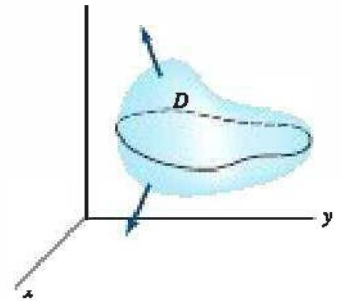


FIGURE 9.16.2 Region D with no vertical side

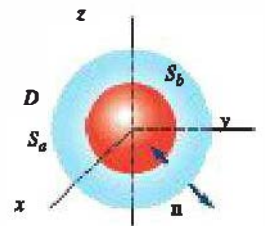


FIGURE 9.16.3 Region D is bounded between two concentric spheres

where $\mathbf{n}$ points outward from D ; that is, $\mathbf{n}$ points away from the origin on S_b and $\mathbf{n}$ points toward the origin on S_a .

EXAMPLE 1 Verifying Divergence Theorem

Let D be the region bounded by the hemisphere $x^2 + y^2 + (z - 1)^2 = 9$, $1 \leq z \leq 4$, and the plane $z = 1$. Verify the divergence theorem if $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + (z - 1)\mathbf{k}$.

SOLUTION The closed region is shown in **FIGURE 9.16.4**.

Triple Integral: Since $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + (z - 1)\mathbf{k}$, we see $\operatorname{div} \mathbf{F} = 3$. Hence,

$$\iiint_D \operatorname{div} \mathbf{F} \, dV = \iiint_D 3 \, dV = 3 \iiint_D dV = 54\pi. \quad (10)$$

In the last calculation, we used the fact that $\iiint_D dV$ gives the volume of the hemisphere ($\frac{2}{3}\pi 3^3$).

Surface Integral: We write $\iint_S = \iint_{S_1} + \iint_{S_2}$, where S_1 is the hemisphere and S_2 is the plane $z = 1$. If S_1 is a level surface of $g(x, y, z) = x^2 + y^2 + (z - 1)^2$, then a unit outer normal is

$$\mathbf{n} = \frac{\nabla g}{\|\nabla g\|} = \frac{x\mathbf{i} + y\mathbf{j} + (z - 1)\mathbf{k}}{\sqrt{x^2 + y^2 + (z - 1)^2}} = \frac{x}{3}\mathbf{i} + \frac{y}{3}\mathbf{j} + \frac{z - 1}{3}\mathbf{k}.$$

Now
$$\mathbf{F} \cdot \mathbf{n} = \frac{x^2}{3} + \frac{y^2}{3} + \frac{(z - 1)^2}{3} = \frac{1}{3}(x^2 + y^2 + (z - 1)^2) = \frac{1}{3} \cdot 9 = 3$$

and so
$$\begin{aligned} \iint_{S_1} (\mathbf{F} \cdot \mathbf{n}) \, dS &= \iint_R (3) \left(\frac{3}{\sqrt{9 - x^2 - y^2}} \, dA \right) \\ &= 9 \int_0^{2\pi} \int_0^3 (9 - r^2)^{-1/2} r \, dr \, d\theta = 54\pi. \quad \leftarrow \text{polar coordinates} \end{aligned}$$

On S_2 , we take $\mathbf{n} = -\mathbf{k}$ so that $\mathbf{F} \cdot \mathbf{n} = -z + 1$. But, since $z = 1$, $\iint_{S_2} (-z + 1) \, dS = 0$. Hence, we see that $\iint_S (\mathbf{F} \cdot \mathbf{n}) \, dS = 54\pi + 0 = 54\pi$ agrees with (10). $\equiv$

EXAMPLE 2 Using Divergence Theorem

If $\mathbf{F} = xy\mathbf{i} + y^2z\mathbf{j} + z^3\mathbf{k}$, evaluate $\iint_S (\mathbf{F} \cdot \mathbf{n}) \, dS$, where S is the unit cube defined by $0 \leq x \leq 1$, $0 \leq y \leq 1$, $0 \leq z \leq 1$.

SOLUTION See Figure 9.13.14 and Problem 38 in Exercises 9.13. Rather than evaluate six surface integrals, we apply the divergence theorem. Since $\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = y + 2yz + 3z^2$, we have from (2)

$$\begin{aligned} \iint_S (\mathbf{F} \cdot \mathbf{n}) \, dS &= \iiint_D (y + 2yz + 3z^2) \, dV \\ &= \int_0^1 \int_0^1 \int_0^1 (y + 2yz + 3z^2) \, dx \, dy \, dz \\ &= \int_0^1 \int_0^1 (y + 2yz + 3z^2) \, dy \, dz \end{aligned}$$

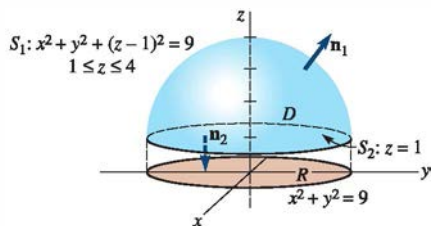


FIGURE 9.16.4 Hemispherical region D in Example 1

$$\begin{aligned}
&= \int_0^1 \left(\frac{y^2}{2} + y^2 z + 3yz^2 \right) dz \\
&= \int_0^1 \left(\frac{1}{2} + z + 3z^2 \right) dz = \left(\frac{1}{2} z + \frac{1}{2} z^2 + z^3 \right) \Big|_0^1 = 2. \quad \equiv
\end{aligned}$$

Physical Interpretation of Divergence In Section 9.14 we saw that we could express the normal component of the curl of a vector field $\mathbf{F}$ at a point as a limit involving the circulation of $\mathbf{F}$. In view of (2), it is possible to interpret the divergence of $\mathbf{F}$ at a point as a limit involving the flux of $\mathbf{F}$. Recall from Section 9.7 that the flux of the velocity field $\mathbf{F}$ of a fluid is the rate of fluid flow—that is, the volume of fluid flowing through a surface per unit time. In Section 9.7 we saw that the divergence of $\mathbf{F}$ is the flux per unit volume. To reinforce this last idea let us suppose $P_0(x_0, y_0, z_0)$ is any point in the fluid and S_r is a small sphere of radius r centered at P_0 . See **FIGURE 9.16.5**. If D_r is the sphere S_r and its interior, then the divergence theorem gives

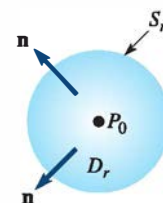


FIGURE 9.16.5 Region D_r in (11)

$$\iint_{S_r} (\mathbf{F} \cdot \mathbf{n}) dS = \iiint_{D_r} \operatorname{div} \mathbf{F} dV. \quad (11)$$

If we take the approximation $\operatorname{div} \mathbf{F}(P) \approx \operatorname{div} \mathbf{F}(P_0)$ at all points $P(x, y, z)$ within the small sphere, then (11) gives

$$\begin{aligned}
\iint_{S_r} (\mathbf{F} \cdot \mathbf{n}) dS &\approx \iiint_{D_r} \operatorname{div} \mathbf{F}(P_0) dV \\
&= \operatorname{div} \mathbf{F}(P_0) \iiint_{D_r} dV \\
&= \operatorname{div} \mathbf{F}(P_0) V_r,
\end{aligned} \quad (12)$$

where V_r is the volume ($\frac{4}{3}\pi r^3$) of the spherical region D_r . By letting $r \rightarrow 0$, we see from (12) that the divergence of $\mathbf{F}$ is the limiting value of the ratio of the flux of $\mathbf{F}$ to the volume of the spherical region:

$$\operatorname{div} \mathbf{F}(P_0) = \lim_{r \rightarrow 0} \frac{1}{V_r} \iint_{S_r} (\mathbf{F} \cdot \mathbf{n}) dS.$$

Hence, divergence $\mathbf{F}$ is flux per unit volume.

The divergence theorem is extremely useful in the derivation of some of the famous equations in electricity and magnetism and hydrodynamics. In the discussion that follows we shall consider an example from the study of fluids.

Continuity Equation At the end of Section 9.7 we mentioned that one interpretation of $\operatorname{div} \mathbf{F}$ was a measure of the rate of change of the density of a fluid at a point. To see why this is so, let us suppose that $\mathbf{F}$ is a velocity field of a fluid and that $\rho(x, y, z, t)$ is the density of the fluid at a point $P(x, y, z)$ at a time t . Let D be the closed region consisting of a sphere S and its interior. We know from Section 9.15 that the total mass m of the fluid in D is given by $m = \iiint_D \rho(x, y, z, t) dV$. The rate at which the mass increases in D is given by

$$\frac{dm}{dt} = \frac{d}{dt} \iiint_D \rho(x, y, z, t) dV = \iiint_D \frac{\partial \rho}{\partial t} dV. \quad (13)$$

Now from Figure 9.7.3 we saw that the volume of fluid flowing through an element of surface area ΔS per unit time is approximated by $(\mathbf{F} \cdot \mathbf{n}) \Delta S$. The mass of the fluid flowing through an element of surface area ΔS per unit time is then $(\rho \mathbf{F} \cdot \mathbf{n}) \Delta S$. If we assume that the change in

mass in D is due only to the flow in and out of D , then the *volume of fluid* flowing out of D per unit time is given by (10) of Section 9.13, $\iint_S (\mathbf{F} \cdot \mathbf{n}) dS$, whereas the *mass of the fluid* flowing out of D per unit time is $\iint_S (\rho \mathbf{F} \cdot \mathbf{n}) dS$. Hence, an alternative expression for the rate at which the mass increases in D is

$$-\iint_S (\rho \mathbf{F} \cdot \mathbf{n}) dS. \quad (14)$$

By the divergence theorem, (14) is the same as

$$-\iiint_D \operatorname{div}(\rho \mathbf{F}) dV. \quad (15)$$

Equating (13) and (15) then yields

$$\iiint_D \frac{\partial \rho}{\partial t} dV = -\iiint_D \operatorname{div}(\rho \mathbf{F}) dV \quad \text{or} \quad \iiint_D \left(\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \mathbf{F}) \right) dV = 0.$$

Since this last result is to hold for every sphere, we obtain the **equation of continuity** for fluid flows:

$$\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \mathbf{F}) = 0. \quad (16)$$

On page 499 we stated that if $\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = 0$, then a fluid is incompressible. This fact follows immediately from (16). If a fluid is incompressible (such as water), then ρ is constant, so consequently $\nabla \cdot (\rho \mathbf{F}) = \rho \nabla \cdot \mathbf{F}$. But in addition $\partial \rho / \partial t = 0$ and so (16) implies $\nabla \cdot \mathbf{F} = 0$.

9.16 Exercises

Answers to selected odd-numbered problems begin on page ANS-24.

In Problems 1 and 2, verify the divergence theorem.

- $\mathbf{F} = xy\mathbf{i} + yz\mathbf{j} + xz\mathbf{k}$; D the region bounded by the unit cube defined by $0 \leq x \leq 1$, $0 \leq y \leq 1$, $0 \leq z \leq 1$
- $\mathbf{F} = 6xy\mathbf{i} + 4yz\mathbf{j} + xe^{-y}\mathbf{k}$; D the region bounded by the three coordinate planes and the plane $x + y + z = 1$

In Problems 3–14, use the divergence theorem to find the outward flux $\iint_S (\mathbf{F} \cdot \mathbf{n}) dS$ of the given vector field $\mathbf{F}$.

- $\mathbf{F} = x^3\mathbf{i} + y^3\mathbf{j} + z^3\mathbf{k}$; D the region bounded by the sphere $x^2 + y^2 + z^2 = a^2$
- $\mathbf{F} = 4x\mathbf{i} + y\mathbf{j} + 4z\mathbf{k}$; D the region bounded by the sphere $x^2 + y^2 + z^2 = 4$
- $\mathbf{F} = y^2\mathbf{i} + xz^3\mathbf{j} + (z - 1)^2\mathbf{k}$; D the region bounded by the cylinder $x^2 + y^2 = 16$ and the planes $z = 1$, $z = 5$
- $\mathbf{F} = x^2\mathbf{i} + 2yz\mathbf{j} + 4z^3\mathbf{k}$; D the region bounded by the parallelepiped defined by $0 \leq x \leq 1$, $0 \leq y \leq 2$, $0 \leq z \leq 3$
- $\mathbf{F} = y^3\mathbf{i} + x^3\mathbf{j} + z^3\mathbf{k}$; D the region bounded within by $z = \sqrt{4 - x^2 - y^2}$, $x^2 + y^2 = 3$, $z = 0$
- $\mathbf{F} = (x^2 + \sin y)\mathbf{i} + z^2\mathbf{j} + xy^3\mathbf{k}$; D the region bounded by $y = x^2$, $z = 9 - y$, $z = 0$
- $\mathbf{F} = (x\mathbf{i} + y\mathbf{j} + z\mathbf{k})/(x^2 + y^2 + z^2)$; D the region bounded by the concentric spheres $x^2 + y^2 + z^2 = a^2$, $x^2 + y^2 + z^2 = b^2$, $b > a$

- $\mathbf{F} = 2yz\mathbf{i} + x^3\mathbf{j} + xy^2\mathbf{k}$; D the region bounded by the ellipsoid $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$
- $\mathbf{F} = 2xz\mathbf{i} + 5y^2\mathbf{j} - z^2\mathbf{k}$; D the region bounded by $z = y$, $z = 4 - y$, $z = 2 - \frac{1}{2}x^2$, $x = 0$, $z = 0$. See **FIGURE 9.16.6**.

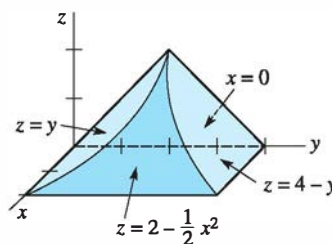


FIGURE 9.16.6 Region D for Problem 11

- $\mathbf{F} = 15x^2y\mathbf{i} + x^2z\mathbf{j} + y^4\mathbf{k}$; D the region bounded by $x + y = 2$, $z = x + y$, $z = 3$, $x = 0$, $y = 0$
- $\mathbf{F} = 3x^2y^2\mathbf{i} + y\mathbf{j} - 6zxy^2\mathbf{k}$; D the region bounded by the paraboloid $z = x^2 + y^2$ and the plane $z = 2y$
- $\mathbf{F} = xy^2\mathbf{i} + x^2y\mathbf{j} + 6\sin x\mathbf{k}$; D the region bounded by the cone $z = \sqrt{x^2 + y^2}$ and the planes $z = 2$, $z = 4$